Enhancing Proactive Emotional Support Systems with Open-Source LLMs: Multi-Agent Architecture, Need-State Memory, and Psychological Evaluation

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I. Project Details

Criterion: Address Limitations and Propose Solutions. We aim to extend the framework of the EMNLP 2025 paper *"Look Beyond Feeling"* (COCOON) [1], identifying architectural and evaluative limitations and proposing targeted enhancements.

Team Composition and Individual Contributions:

- *Shared work:* Data construction pipeline the corpus on which every later enhancement depends, fine-tuning of Open Source LLM and turn-level psychological metrics.
- **Arpit Bhomia 12340340**, email: arpitba@iitbhilai.ac.in Responsible for the Temporal dynamics, need-state memory.
- **Ashish Ranjan 12340370**, email: ashishr@iitbhilai.ac.in Responsible for the Inference architecture, multi-agent architecture.

II. Problem Statement

Context and Motivation

Mental health disorders affect nearly one billion people globally, yet most never receive treatment — over 75% in low- and middle-income countries, by WHO estimates. The barrier is often not the absence of support but the nature of distress itself: people in crisis suppress feelings, avoid therapy, and resist direct questioning. Even where services exist, stigma, denial, and *reactance* — the tendency to resist perceived threats to autonomy — prevent many from articulating their needs.

Existing AI-driven emotional support systems are overwhelmingly *reactive*: they wait for the user to explicitly state a problem before offering help. This design assumption fails the very population most in need — those who will not or cannot ask. The challenge is to **proactively** initiate and sustain dialogue, uncovering latent needs from implicit cues *without* triggering the resistance that prevents help-seeking in the first place.

The problem sits at the intersection of *affective computing*, which must infer emotional states from ambiguous signals; *dialogue policy learning*, which must decide when and how deeply to probe; and *psychologically grounded evaluation*, which must measure outcomes against established psychological constructs rather than surface fluency. Because a poorly timed question can itself cause withdrawal, the domain is a demanding testbed for responsible AI.

What COCOON Did

COCOON [1] introduced a proactive support model grounded in active listening, trained on a corpus the authors state is written in Chinese: rather than asking direct questions, it maps surface *Feelings* to underlying *Needs* to generate responses that feel natural and non-intrusive. Strategies are annotated at the *content level* — free-form phrase summaries — rather than from a fixed atomic taxonomy, which the authors identify as the key advantage over ESConv [2] and ExTES [3]. The authors' own Limitations section names conversation length and the absence of extended, continuing interaction as open problems. Neither the corpus nor the pipeline that produced it — profile construction, dialogue generation, filtering, and annotation — is available, so our enhancements aim to extend limitations the authors partly acknowledge, and would need to rebuild the construction pipeline in English before extending it. Fig. 1 illustrates COCOON's core pipeline and the three architectural gaps we seek to address.

Specific Scope of Our Work

The framework's user model is static and single-session, its inference is monolithic, and its central claim about conversational resistance is validated only in aggregate. We propose four modular enhancements, each designed to be independently testable:

(1) A data construction pipeline intended to yield the first *proactive* English corpus with multi-session temporal structure.
(2) A multi-agent inference pipeline with evidence-grounded contracts and a conformally gated safety critic.
(3) Turn-level psychological metrics (IP and PRI) designed to localize resistance and feed back into generation.
(4) A need-state memory purposed to persist inferred need chains as revisable beliefs across sessions.

Each enhancement maps directly to a gap in Fig. 1, and we aim to compose them to yield the full pipeline shown later in Fig. 3.

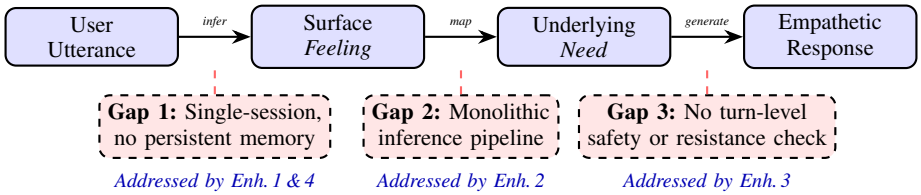


Fig. 1. COCOON's Feeling→Need→Response pipeline (solid blue) and the three architectural gaps (dashed red) our proposed enhancements intend to address. Gap 1: no cross-session memory or temporal modelling. Gap 2: all reasoning collapsed into a single forward pass with no intermediate inspection point. Gap 3: no per-turn safety gate or resistance measurement to prevent over-probing.

III. Methodology

Our approach builds on the COCOON framework. We aim for each enhancement to be modular — and we intend to choose every component to fit within the memory budget of a single GPU using openly available models.

A. Enhancement 1: Data Construction with Temporal Structure

We plan to reimplement a data construction pipeline from published methodologies — profile construction over the emotion, feeling, need and memory dimensions with its bounded need-inference chain, two-stage dialogue generation covering the listening and suggestion phases, quality filtering using the Active-Empathic Listening Scale, and fine-grained turn annotation producing content-level Analysis and Strategy labels. We aim to release this pipeline so that the corpus can be regenerated, extended, or rebuilt at a different scale.

Profile seeding. The original profiles are fully LLM-synthesized [1]: a handful of hand-written seed examples are placed in the prompt and every situation is then sampled from the generator. We aim instead to seed situations from EmpatheticDialogues [4] (human-written conversational narratives rather than help requests) and to use ESConv for problem-type stratification. We treat this as an attempt rather than a settled design: many of those situations are short and may not sustain a three-step need chain, so if the surviving yield after filtering is too small or the resulting profiles do not look promising, we would instead build them fully LLM-synthesized. We intend to report which of the two routes produced the released corpus. Proactivity is intended to be injected at role-play time: the simulated user did not seek help and should initially resist disclosure. Our stratified persona attempts to pair observable surface traits with a hidden concern the supporter must uncover, inspired by [5] but adapted for emotional support — the hidden layer is meant to be the terminal node of the need chain together with a resistance level that the supporter's behaviour can raise or lower.

Multi-session temporal markers. The original setup models isolated single sessions. We intend to generate linked session sequences separated by explicit time gaps: an elapsed interval is sampled and injected into each follow-up session context, and the profile is advanced accordingly — needs partially resolve, others intensify, new events accumulate, earlier concerns decay in salience. The English speaker prompt in COCOON's public repository is the *reactive* confiding role and omits the need field; only the proactive, resistance-conditioned role-play prompt is released, and it is Chinese, so no English proactive corpus follows from the released artefacts. An English multi-session support corpus has since been released [6], but its users are help-seekers with no supporter annotation; we aim to condition ours to resist, the only setting in which intrusiveness and reactance are meaningful. These intended time jumps make every remaining enhancement testable: they aim to give memory meaningful timestamps, create recency-versus-relevance tension, and enable cross-session recall measurement. Fig. 2 illustrates the planned linked session structure.

B. Enhancement 2: Multi-Agent Architecture

We propose decomposing the monolithic listener into specialized agents (Analyzer, Strategist, Generator, Critic). We do not claim decomposition itself as novel: multi-agent role-play and chain-of-thought analysis-plus-strategy supervision are both established in emotional support, and a recent counseling framework already splits a strategy module from a generation module [7]. Decomposition is infrastructure here; our intended contribution is what the boundary it creates is used for — the evidence-grounded write contract and the conformally gated critic described below. We would prefer to run this 4-agent pipeline concurrently, but would adapt the execution strategy (e.g., orchestrating them sequentially or utilizing model offloading) in accordance with our strict hardware constraints. The **Analyzer** is designed to ingest dialogue history and emit a structured psychological state (emotional state, implicit needs, resistance level) where every inference is expected to carry supporting quotes and a confidence score. The **Strategist** aims to produce a content-level strategy plan with a justification chain; where analyzer confidence is low, strategies are meant to be restricted to exploratory moves. The **Generator** is tasked with drafting the utterance. Finally, the **Critic** intends to score the draft on Enhancement 3 metrics, returning structured feedback for revision or falling back to a plain reflective statement if revision fails.

Two proposed rules govern the contract: (1) structured output *augments* rather than replaces raw dialogue history, so every agent aims to see the original user words; (2) the critic is intended to operate under a *conformal risk gate* — instead of a tuned confidence threshold (whose calibration does not transfer across dialogue distributions), we plan to expose an intrusiveness error budget α, calibrate a nonconformity score from the Intrusiveness Penalty on a held-out split, and obtain a distribution-free bound on the fraction of released turns exceeding the tolerance. The grounding contract adapts

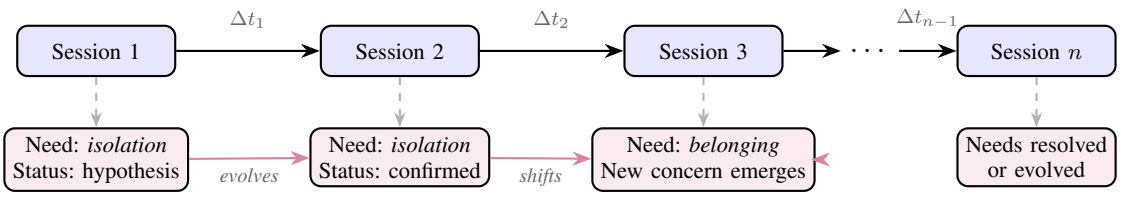


Fig. 2. Multi-session temporal structure. Linked sessions are separated by sampled time gaps (Δt_i). The user profile evolves between sessions: needs may be confirmed, resolved, or replaced by new concerns. This structure aims to enable meaningful evaluation of cross-session memory and need-state tracking across the full trajectory of emotional support.

typed grounding from multi-agent hallucination detection, but proposes a decisive category — an inference the user has *denied* — as the highest-weight gate signal, since this is the textbook signature of premature interpretation.

Scope of the guarantee. Conformal coverage requires calibration and deployment turns to be exchangeable, so the bound we can claim holds only on held-out turns drawn from the same generator and profile distribution as the calibration split. The cross-corpus arms of Sec. III are deliberately out of distribution; there we plan to report the empirical violation rate as a measured quantity, not as a guarantee, and to treat any gap between the two as evidence about how far the gate transfers.

A 2×2 ablation. Because decomposition is infrastructure and the gate is the claim, a single decomposition ablation cannot say which factor pays. We therefore plan to cross the two — decomposed versus monolithic listener, gated versus ungated critic — and report all four cells on Success Rate and Intrusiveness Penalty. The informative outcome is not a joint win. If the gate carries the effect while decomposition alone contributes nothing, that is what we would report, and it would be a stronger statement about where safety in these pipelines comes from than any aggregate improvement would be.

C. Enhancement 3: Turn-Level Psychological Metrics

We aim to implement two judge-based metrics operating at *turn* granularity, operationalizing constructs from the psychological literature:

- **Intrusiveness Penalty (IP):** Intended to score each supporter turn on whether its insight oversteps user readiness — premature advice-giving, asserting undisclosed emotional states, and boundary violations. Items derive from the clinical notion of premature interpretation and privacy-management accounts of disclosure.
- **Psychological Reactance Index (PRI):** Designed to score the user's *following turn* for resistance aroused by the preceding supporter move, following the operationalization of reactance as the conjunction of anger and negative cognition [8]. In dialogue this often surfaces as deflection, topic changes, shortened responses, and increased defensiveness. Because PRI aims to read the actual next turn, it seeks to measure realized resistance.

Counterfactual attribution. To separate supporter-provoked from pre-existing resistance, we propose treating each supporter turn as an intervention: score the exchange as-is, then replace the supporter turn with a neutral reflective control and re-generate the user turn. Writing r_1 for the score of the factual exchange and r_0 for that of the neutral control, $\mathrm{PRI} = \bar{r}_1 - \bar{r}_0$ across N rollouts per condition, reported with a confidence interval, following the counterfactual-replay style of attribution developed for LLM-agent failures. Recent work [7] measures turn-level resistance as Resistance Triggering Frequency, but scores only the client side and is observational; our proposed IP aims to score the supporter, while PRI seeks to be graded with explicit causal attribution.

What PRI can and cannot claim. Enhancement 1 scripts a resistance level into the profile, and the same model family both plays the user and scores the turn. A supporter could therefore lower PRI by matching the script rather than by being less intrusive, and the judge could reward its own generation style. We plan three controls: the resistance level would be withheld from the judge prompt; the judge would be a different open model from the one used for generation and role-play, which matters more here than in the original setup because PRI scores the simulator's own turn; and a stratified subsample would be rated by the team against the same rubric, so that judge–human agreement can be reported alongside every PRI number. We would treat PRI as resistance realized *by a simulated user*, and intend to state transfer to human reactance as out of scope.

D. Enhancement 4: Need-State Trajectory Memory

Rather than a store of facts, we aim to persist the system's *belief about the user*. The intended unit of memory is COCOON's own construct — the need chain from surface feeling to terminal need — planned to be kept as a revisable object rather than re-inferred each session. The idea is inspired by belief tracking in statistical dialogue management, where the system carries a hypothesis over an unobservable user goal and revises it from evidence instead of committing to one parse; what we propose as new here is that the hypothesis is an emotional need, that it survives across sessions, and that it aims to record disconfirmation as a first-class state.

Each chain node is designed to carry a status (*hypothesis*, *confirmed*, *disconfirmed*) with the user spans supporting or contradicting it, their timestamp, and a confidence value; the Analyzer's proposed grounding contract also aims to govern writes, so that nothing enters the belief state ungrounded. We plan for follow-up sessions to revise rather than append: a need acted on is intended to be marked resolved, an inference the user denied is marked disconfirmed and blocked from re-proposal, and a concern resurfacing after a long gap is

reinstated at reduced confidence, with the intervals of Enhancement 1 setting decay on unconfirmed hypotheses. We envision retrieval as a question, not a lookup: the memory is meant to report the current best hypothesis about the terminal need and which evidence is still outstanding, so what surfaces is driven by what the system does not yet know rather than by similarity to the latest turn.

Approach. This is deliberately proposed as not a general-purpose graph memory. Temporal knowledge graphs are standard infrastructure by now, and a strong recent baseline shows structurally simple event-level memory matching or surpassing elaborate graph retrieval on long-horizon benchmarks [9]; another triplet store would likely spend the resources on a component whose marginal value is already contested. Our hypothesis is that the useful structure in emotional support is not the entity graph but the *inference chain*, and that what memory must try to carry forward is which links have been tested.

This is intended to target the conflict-detection and abstention failures current benchmarks report [6]. A system with no record of what it already got wrong cannot easily detect a conflict or stop re-proposing a rejected inference — that is the mechanism we aim to supply, and the reason we plan to track disconfirmation explicitly rather than only recency and valence. We propose comparing against session-isolated baseline, flat running-summary memory, dense retrieval, and the event-level baseline of [9], aiming to report **Success Rate** (need identification) rather than merely recall.

E. Fine-Tuning on Open-Source Models

Every model considered for this project is openly available; we plan to avoid any proprietary API at any stage. We intend to obtain the supporter by supervised fine-tuning of an open instruction-tuned base model on the proposed corpus of Enhancement 1 with per-turn Analysis and Strategy annotations in the target sequence, so intermediate reasoning is meant to be learned rather than prompted. We plan to conduct training using 4-bit quantized low-rank adapters with gradient checkpointing, aiming for feasibility on a single GPU. An intended *w/o thoughts* arm trained without annotations seeks to isolate their contribution and serve as our proposed reproduction check against the published ablation.

F. Evaluation Protocol and Baseline Comparison

Neither the COCOON corpus nor its construction pipeline is available, and the original work is in Chinese while ours is in English, so a number-for-number comparison against their published table is neither available nor meaningful. We therefore aim to publish results from our fine-tuned open-source model as standalone benchmarks. Comparability rests on the instruments rather than on shared code: AELS, the Comforting Responses Scale, and the RAC Scale are published psychological instruments, and we intend to implement them ourselves for English with open judge models, and to report the item wording we use so that any difference from the original scoring stays visible rather than assumed away. We plan to evaluate on our regenerated reticent profiles, on the ExTES user profiles [3], and on ES-MemEval [6], so that memory is tested on data we did not generate. We intend to report Success Rate (fraction of dialogues identifying the underlying need), the basic metrics, psychological scale dimensions, and our proposed turn-level IP/PRI metrics. Reporting Success Rate alongside IP and PRI forms our key intended test: a system might raise Success Rate by interrogating the user, and we hope the resistance metrics will reveal if it did so at the cost of intrusiveness. Because our corpus, our user simulator, and our judge would all be open models, we intend to fix one further rule: generation and role-play may share a model, as they did in the original work, but the judge would be neither, and we aim to report every headline number with the judge–human agreement measured on its stratified subsample.

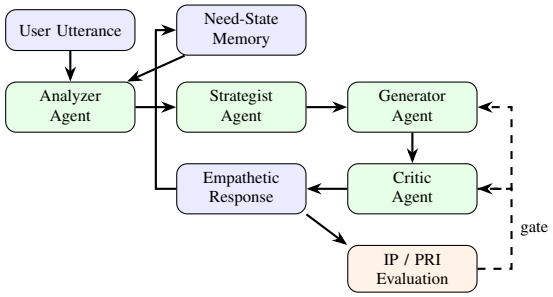


Fig. 3. Proposed system pipeline. Turn-level metrics are intended to gate the critic, and responses update the need-state memory for cross-session continuity.

IV. Resource and Dataset Availability

A. Dataset

- **Source corpora:** All openly downloadable. Empathetic-Dialogues [4] provides human-written situations narrated conversationally and is proposed as our primary profile seed; ESConv [2] contributes a problem-type taxonomy for topical stratification and a reactive baseline; ExTES [3] serves as the strongest reactive baseline and our intended second evaluation profile set.
- **Original corpus:** The COCOON corpus is not released, nor is the profile-construction, two-stage generation, filtering, or annotation pipeline that produced it. The original work is also Chinese, whereas ours is English, so its artefacts would not transfer directly even if released. We therefore plan to build the construction pipeline and all role-play prompts ourselves in English from the published methodology, and to implement the evaluation scales from their source psychological literature rather than inheriting a scoring implementation. COCOON remains the only *proactive* support corpus; multi-session corpora released since [6] are reactive.
- **Our proposed corpus:** English dialogues intended to have full user profiles, content-level per-turn Analysis and Strategy annotation, and linked multi-session sequences with explicit time gaps. Final scale is expected to follow the yield of generation and filtering.

B. Resources

- **Hardware:** We intend to attempt our quantized fine-tuning and serving using a single 12 GB GPU (such as an NVIDIA RTX 3060). A 12 GB budget would not hold four agents, a user simulator, and a larger judge concurrently, so the plan is serial: one model resident at a time, with the four agent roles sharing a single quantized base model switched by role prompt and adapter, and corpus generation and judging run as offline batch passes rather than alongside inference. Only two distinct models are needed across the whole schedule — the shared generation and role-play model, and the judge — and they would never be resident together. We acknowledge that additional VRAM or a multi-GPU configuration may become necessary, and intend to report which components were run at reduced scale if so.
- **Software:** Python, PyTorch, HuggingFace Transformers, parameter-efficient and quantized training libraries (PEFT, bitsandbytes), a batched inference server, and a conformal prediction library for the proposed critic gate.
- **Models:** We plan to use exclusively open weights, and to keep the roles on two distinct models rather than three. One instruction-tuned model would serve both corpus generation and user role-play, as in the original setup. A second, larger open model from a different family would serve as judge, so that it is distinct from the fine-tuned supporter it scores — the configuration the original work already used, since their judge scored a fine-tuned Llama supporter — and, additionally, distinct from the user simulator. The extra separation is required by PRI rather than inherited: PRI reads the simulated user's next turn, so a shared judge and simulator would have a model rating its own output. An open base model would be fine-tuned to obtain the supporter.

V. Limitations Found

Three limitations define our intended contribution, each paired with its proposed solution. Limitation 3 is one the authors partly acknowledge in their own Limitations section; Limitations 1 and 2 are ours. We state this explicitly because extending an acknowledged limitation is a different claim from finding an unnoticed one:

- **Limitation 1 — Monolithic Cognitive Overload.** One model analyzes psychology, selects a strategy, and generates the response in a single forward pass, limiting reasoning depth per subtask and leaving no boundary at which a safety check could intervene.
  Proposed Solution: Enhancement 2 aims to decompose the pipeline into specialized agents with evidence-grounded contracts, trying to create a boundary at which a critic can inspect and revise.
- **Limitation 2 — Coarse-Grained Resistance Validation.** The central claim that direct probing induces resistance is tested only in aggregate by steering a global questioning ratio. No metric attributes resistance to an individual supporter turn, so the framework can neither identify which move caused a withdrawal nor act on it.
  Proposed Solution: Enhancement 3. We intend for turn-level metrics to localize resistance to the utterance that provoked it and feed it back into generation through the critic as a proposed safety gate.
- **Limitation 3 — Context Collapse Across Sessions.** Dialogues average roughly nineteen turns but are isolated episodes; no state is retained between them. States that evolve, partially resolve, or resurface cannot be represented, a gap the authors themselves flag when they note that real support requires continuing interaction.
  Proposed Solution: Enhancements 1 and 4. Time gaps aim to produce linked sessions in which the user's state genuinely changes, and we propose a need-state memory to persist the need chain as a revisable belief recording which inferences were confirmed and which denied.

What aims to differentiate our work:

- The intention to build the first *proactive* multi-session English support corpus with content-level supporter annotation; concurrent corpora are reactive and unannotated on the supporter side.
- The proposed Intrusiveness Penalty, aimed as a supporter-turn measure of premature interpretation, and a reactance index we plan to attribute by intervention rather than by frequency; recent turn-level work scores only the seeker.
- A proposed conformal risk gate on the critic in place of a tuned threshold, and an intended grounding contract whose decisive category — an inference the user has denied — has no counterpart in factual grounding.
- A plan to handle memory as a maintained belief over the need chain rather than a store of facts, aiming to target the conflict-detection and abstention failures current benchmarks report.

VI. Expected Outcomes

By the conclusion of this project, we aim to deliver:

1) **Pipeline and corpus:** A runnable intended reimplementation of the proactive user simulator, and the linked multi-session corpus we aim for it to produce. Targeted acceptance criterion: aim to reproduce the direction of the published annotation ablation and attempt to beat reactive baselines on Success Rate.
2) **Refactored system and ablation:** A proposed modular multi-agent pipeline aiming to replace the monolithic listener, compared like-for-like on the psychological scales, with a 2×2 ablation crossing decomposition against the critic gate so that the contribution of each factor is separable rather than confounded.
3) **Validated metrics:** Intended turn-level IP/PRI scores for baselines and our system, a planned judge reliability check, and the aim to map the critic's recovery-versus-cost curve.
4) **Persistent memory:** A proposed need-state memory we aim to evaluate on our sequences and ES-MemEval [6], against flat-summary, dense-retrieval, and event-level [9] baselines, aiming to measure Success Rate as well as recall.

References

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APPENDIX

To facilitate reproducibility, we aim to release all prompts, model configurations, quantization settings, and hyperparameters alongside our codebase. Should any configuration exceed our hardware budget—potentially necessitating adjustments to the model scale, batch size, or number of rollouts—we intend to fully document these substitutions along with their underlying rationale. Because the original work is Chinese and ours is English, the pipeline, the role-play prompts, and the evaluation scales are to be written by us — from the published methodology and the underlying psychological instruments respectively, not inherited from an existing implementation. Automated drafting tools assisted in structuring and refining this text. As a preliminary framework, this document omits final metrics, benchmarks, and results; we aim to compile and deliver those later in the course. Because this is an active project, our actual contributions may shift and will be formally detailed in later evaluation phases.